Staged training and long-context ladders

> **Reading note.** This is an explanatory reconstruction of Jiaxuan Zou's literature synthesis. Published experiments belong to the cited teams. Numbered protocols and illustrative matrices are our planning synthesis unless explicitly attributed.

The blog's §11 extends the basic ladder to training that changes distribution or context length. A final model depends on the path taken through those stages, so an endpoint fit for a single fixed distribution may miss the delivered model's behavior.

Staged training

1. Define the backbone pretraining, higher-quality continuation, context extension, and wrap-up stages actually planned for delivery.
2. Specify token fractions, mixture weights, schedule state, and transitions for each stage.
3. Use shared checkpoints to compare continuation candidates efficiently, recording common-prefix dependence.
4. Include the complete staged path in larger confirmations. A continuation-only win does not establish that the same mixture is best throughout pretraining.
5. If post-training is included, apply a standardized small post-training probe to candidate checkpoints.

For stage s with D_s tokens and bucket weight w_{js}, cumulative bucket exposure is

$$e_j = \frac{\sum_s D_s w_{js}}{U_j}.$$

This makes clear why resetting epoch counters at stage boundaries hides repeated use of the same data. Recompute the denominator if unique pools actually change.

Long-context training

Longer sequences change attention compute, token packing, memory, batch feasibility, and positional behavior. Test context extension separately from a new optimizer or mixture whenever possible.

1. Choose target lengths and positional settings, then recompute FLOPs and feasible token batches.
2. Measure loss across context positions and document lengths.
3. Evaluate retrieval, multi-hop aggregation, and realistic long-document tasks rather than relying on nominal context capacity.
4. Check that short-context capabilities survive extension.
5. Repeat at a larger scale to test transfer of the extension recipe.

Deliverable: a stage manifest and context-evaluation matrix, with cumulative token and data-exposure accounting.

Sources: blog §11; OLMo 2, 2025; OLMo 3, 2025; RULER, 2024.

Blog/notes crosswalk · Detailed companion